

# Solubility in Water

## Lesson objectives

- Explain how concentration affects dissolving or saturation.
- Investigate the solubilities of different solutes in water
- Be able to identify when a solution is saturated

# Solubility in Water

1. What is a solute?

**Something that can dissolve in a solvent**

2. What is a solvent?

**Something that a solute dissolves into**

3. If you have a concentrated solution and add water to it, describe the change in the solution

(keywords to use in your answer: dilute, concentrated)

**It would become more dilute/ less concentrated**

4. Describe the difference between dissolving and melting

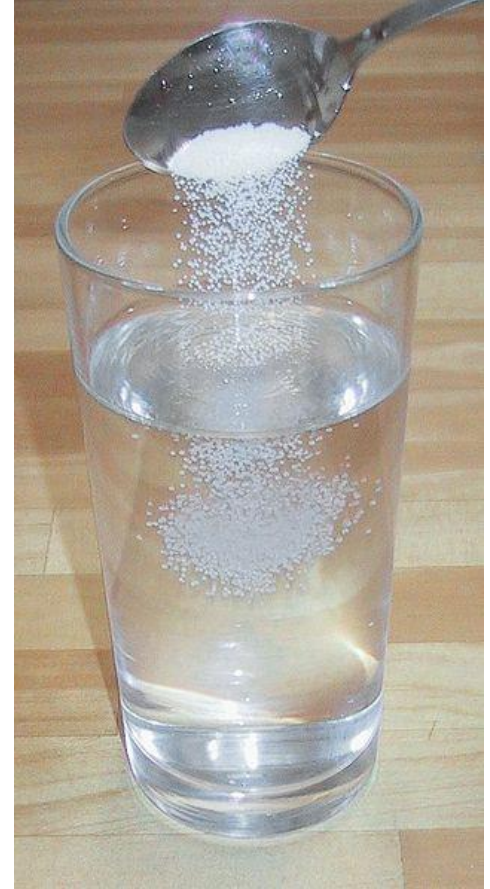
**Dissolving is when a solute dissolves into a solvent, melting is a change of state from solid to liquid due to temperature**

## Solubility in Water

What would happen if i continue to add more and more sugar?

The **concentration** of the solute would increase

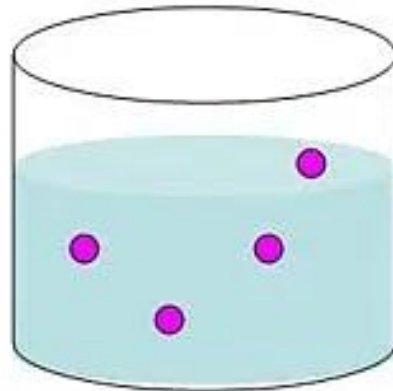
Eventually the sugar will stop dissolving



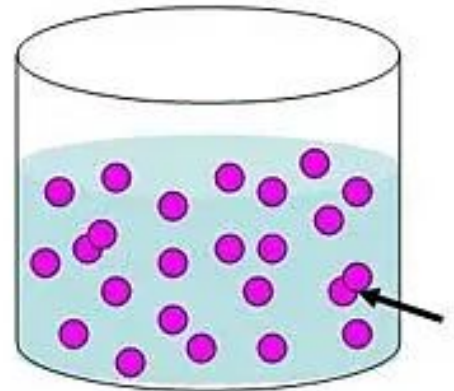
# Solubility in Water

A solution that **can** still dissolve more solute is **unsaturated**

A solution that **cannot** dissolve any more solute is **saturated**



Unsaturated solution

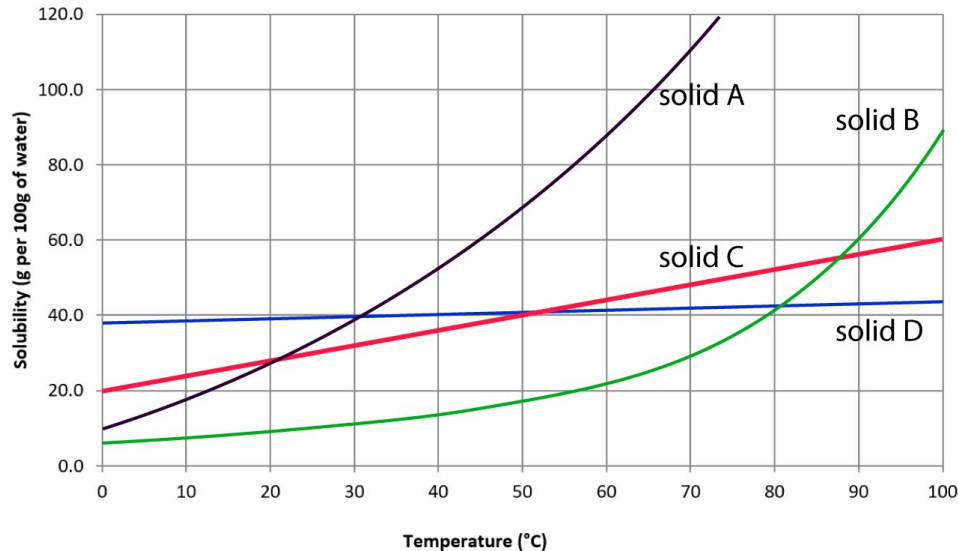


Saturated solution

# Solubility in Water

The amount of solute that is able to dissolve is the **solubility** of that solute.

Different solutes have different solubility in water.



We will look at temperature next lesson

# Solubility in Water

You are going to investigate the solubility of 3 different solutes in water

Pg 48

Think like a scientist

## Solubility in water

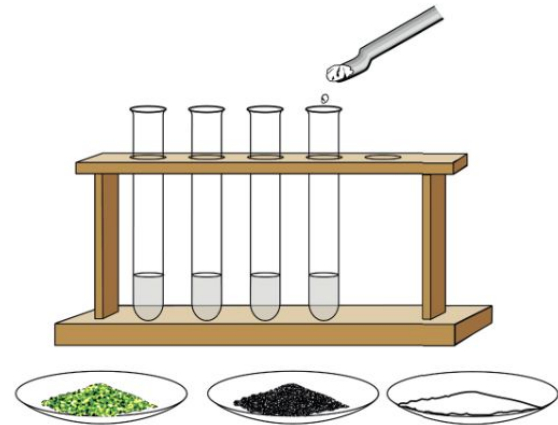
You will need:

- test tubes
- test tube rack
- safety glasses
- measuring cylinder
- spatula
- a range of solutes such as sodium chloride, potassium sulfate and sugar

In this task you will use various solutes and investigate their solubility in water. You will use water at room temperature.

## Method

- 1 Place a measured volume of water in a number of test tubes. Use a different test tube for each of the solutes.
- 2 Add the first solute to the water. Count how many spatulas of the solute you can add until no more will dissolve. After you add each spatula of the solute, shake or stir the contents of the test tube carefully.
- 3 Repeat for the other solutes.
- 4 Record your results in a table.



# Solubility in Water

1. What are we investigating?

**Solubility of different solutes**

2. What is the independent variable in this investigation? (what you change)

**The type of solute**

3. What is the dependent variable in this investigation? (what you measure)

**The number of spatulas of solute added**

4. What is a control variable in this investigation? (keep the same)

**Volume of water, the temperature, size of spatula**

# Solubility in Water

## Questions

- 1 Which was the most soluble of the solutes you used?
- 2 Which was the least soluble of the solutes you used?
- 3 In this investigation you used the number of spatulas as a measure of the quantity of solute added. Suggest another way of measuring the quantity of solute used, to improve the accuracy of the results.

## Stretch:

Can you draw a particle diagram to show the solute dissolved in the solvent for your different solutions

# Solubility in Water

Let's practice...

**Workbook Pg 21 - 26**

Exercise 2.1B and 2.1C pg 22

Exercise 2.2A and 2.2B pg 24



# Topic 2.1 Dissolving

## Exercise 2.1A Using the correct scientific term

- A mixture
- B filtrate
- C solute or accept solid
- D solvent
- E solution

Mark your work using a green pen!

## Exercise 2.1B What is the difference between these terms?

- 1** If something is transparent, you can see through it. For example: clear plastic or a solution of copper sulfate is transparent and you can see what is inside or behind it. Something that is opaque does not allow light through it, for example, milk or a container made of ceramic or metal does not allow you to see what is inside.
- 2** When something dissolves, such as sugar in tea, there are two substances involved: the sugar that dissolves and the tea that it dissolves into. When something melts, only that one substance is involved: ice cream on a hot day, for example.
- 3** A solute is something that dissolves in a solvent to form a solution.

**Mark your work using a green pen!**

## Exercise 2.1C Explaining observations

- 1 120 g
- 2 When the salt dissolves in the water, it does not disappear, but is still in the water, although you can't see it any more. You have added 20 g of salt to 100 g of water so there will be 120 g of solution.

## Topic 2.2 Solutions and solubility

### Exercise 2.2A Using the correct scientific term

- 1 insoluble
- 2 concentrated
- 3 saturated solution
- 4 soluble
- 5 diluted

**Mark your work using a green pen!**

## Exercise 2.2B Looking at the solubility of three solutes

- 1 Solute A
- 2 Solute B
- 3 Yes, Arun's results do agree with Zara's: he has the solubility of the solutes in the same order as her.
- 4 Arun used more water than Zara in his investigation, so he was able to dissolve more solute.
- 5 Marcus has not used the same volume of water for each of his tests. He used about the same volume of water for solute A as Zara did and he got the same result as she did. For solute B, he used more water than Zara and about the same volume as Arun and got the same result as him. For solute C, Marcus used more water than Zara but less than Arun and his result was different from both of theirs. Marcus should have used the same volume of water when he tested each of the solutes. He cannot compare the solubility of the three solutes because he did not make his test fair. He changed two things, the solute he was using and the volume of water.

**Mark your work using a green pen!**